

**PROGRAMME: BCA (Bachelor of Computer Applications)
SEMESTER – I**

Teaching-Learning & Evaluation Plan

Course Information: Core Course

Course Code: **23BCA1C04**

Course Title: **Programming in C**

Credits Units: **03**

Contact Hours: **45**

L-T-P: **3-0-0**

CA: UE Weightage - **50:50**

Pass Marks (CA & UE) – **20:20**

Aggregate Pass Marks: **40**

UE Question Paper Marks: **50**

Special Examination Fees: **NA**

Prerequisite (if any): **NA**

Course Facilitator (s): Dr.S.Pandikumar, Dr. N.R Solomon Jebaraj, Dr. Shyam and Ms. Smita, Dr.Deepak Mekhta

Programme Outcomes (POs)	
At the end of the programme, students will be able to	
PO 1	Computational Knowledge: Understand and apply mathematical foundation, computing and domain knowledge for the conceptualization of computing models from defined problems.
PO 2	Problem Analysis: Ability to identify, critically analyze and formulate complex computing problems using fundamentals of computer science and application domains.
PO 3	Design / Development of Solutions: Ability to transform complex business scenarios and contemporary issues into problems, investigate, understand, and propose integrated solutions using emerging technologies.
PO 4	Conduct Investigations of Complex Computing Problems: Ability to devise and conduct experiments, interpret data and provide well-informed conclusions.
PO 5	Modern Tool Usage: Ability to select modern computing tools, skills and techniques necessary for innovative software solutions.
PO 6	Professional Ethics: Ability to apply and commit professional ethics and cyber regulations in a global economic environment.
PO 7	Life-long Learning: Recognize the need for and develop the ability to engage in continuous learning as a Computing professional.
PO 8	Project Management: Ability to understand management and computing principles with computing knowledge to manage projects in multidisciplinary environments.
PO 9	Communication Efficacy: Communicate effectively with the computing community as well as society by being able to comprehend effective documentations and presentations.
PO 10	Societal & Environmental Concern: Ability to recognize economic, environmental, social, health, legal, ethical issues involved in the use of computer technology and other consequential responsibilities relevant to professional practice.
PO11	Individual & Team Work: Ability to work as a member or leader in diverse teams in a multidisciplinary environment.
PO12	Innovation and Entrepreneurship: Identify opportunities, entrepreneurship vision and use of innovative ideas to create value and wealth for the betterment of the individual and society.
Program Specific Outcomes (PSO's)	
PSO 1	Pertain current knowledge and adapting to emerging applications of Mathematics, Science fundamentals in the field of Computer science and its applications.

PSO 2	Exhibit proficiency in identifying, formulating and analyzing complex problems in the computer environment.
PSO 3	Ability to create, select and apply appropriate modern techniques for solving complex issues.
PSO 4	Explore technical knowledge in diverse areas of Computer Applications and experience a conducive environment in nurturing skills for successful career and higher studies.

Course Objectives (COB's)

COB1: To understand the fundamental programming concepts and methodologies which are essential to building good C programs.

COB2: To apply the fundamental programming methodologies in the C programming language via laboratory experiences.

COB3: To create a well-structured, reusable, robust computer program using the C programming language.

Course Outcomes (CO's)

At the end of the course, students will be able to

CO1: Understanding problem design, implementation, primitive data types, values, operators and expressions in C

CO2: Apply basic data structures and control flow for solving required tasks

CO3: Develop modular program development with the help of functions and use it effectively for problem solving

CO4: Implement the concepts of pointers and structures based on the problem domain

CO5: Demonstrate file handling operations and preprocessor directives

Sl. No.	Course Outcome	Skill's	PO's	Bloom's Taxonomy Level	Assessment Tool
1.	CO 1	Understand	PO1,PO2,PO3,PO5,PO7	L2	Problem Solving
2.	CO 2	Apply	PO1,PO2,PO3,PO5,PO7,P O9,PO11,PO12	L3	Problem Solving, Mini Project
3.	CO 3	Develop	PO1,PO2,PO3,PO5,PO7,P O9,PO11,PO12	L6	Problem Solving, Mini Project
4.	CO 4	Implement	PO1,PO2,PO3,PO5,PO7,P O9,PO11,PO12	L3	Problem Solving, Mini Project
5.	CO 5	Demonstrate	PO2,PO3,PO7,PO9,PO11, PO12	L3	Presentation

CO-PO/PSO Mapping:

Course Outcome	Blooms Taxonomy Levels	Program Outcomes (PO's) Mapped	Program Specific Outcomes (PSO's)
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		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
														L3	L4	L6	L4
CO1	L2	2	2	3	-	1	-	2	-	-	-	-	-	3	2	1	2
CO2	L3	2	1	3	-	2	-	2	-	2	-	2	2	2	1	3	2
CO3	L6	2	1	3	-	1	-	2	-	2	-	2	2	2	1	3	2
CO4	L3	1	2	2	-	2	-	1	-	2	-	2	2	-	1	2	1
CO5	L3	-	1	2	-	-	-	1	-	2	-	2	2	-	1	2	1

Course Contents:

Module	Details	Contact Hours
I	Introduction to computer - based problem solving , Program design and implementation issues. Prepare flowcharts & Develop Algorithms, Top-down design & stepwise refinement, Programming environment, Machine language, assembly language, high level languages, Assemblers, Compilers, Interpreters. Overview of C , Data Types, Type modifiers and storage class specifiers for data types, Constants & Variables, Operators, ?Operator, &operator, * operator, & Expressions	9
II	Fundamentals of C programming: Branching and Looping - IF statement, IF..else statement, Nesting of if..Else statement, Else if ladder, Switch Statement, The? : Operator, Go To statement, While, Do while, For, break and continue statements. Arrays - One, Two & multidimensional arrays. Strings – Introduction to strings. Advanced features -Type casting, type conversion. Formatted and unformatted Input/Output - scanf(), printf(), gets(), puts(), getch() and putch().	10
III	Functions -Introduction to Library Functions, Introduction to User defined functions, Call by value and Call by reference, calling functions with arrays, argc and argv, Recursion - basic concepts, ex-towers of Hanoi.	8
IV	Dynamic data structures in C: Pointers - Introduction to pointers, Declaration and Initialization, The & and * operator, pointer expression, arithmetic, comparison, malloc vs calloc, arrays of pointers, pointers to pointers, pointers to functions, function returning pointers. Structures - Basics, declaring, referencing structure elements, array of structures, passing structures to functions, structure pointers, arrays and structures within structures. Unions – Declaration, uses, enumerated data-types, typedef	12
V	Additional features: File Handling – The file pointer, file accessing functions, fopen, fclose, puc, getc, fprintf. C Preprocessor - #define, #include, #undef, Conditional compilation directives. C standard library and header files: Header files, string functions, mathematical functions, Date and Time functions.	6

Text Books (TB):

TB-1	Programming in ANSI C by Balaguruswamy, 6th Edition, 2017, Tata McGraw Hill
TB-2	Let us C by Yashwant Kanetkar, 6th Edition, PBP Publication.
TB-3	The C programming Language by Richie and Kenninghan, second edition, 2012, BPB Publication.

Reference Books (RB):

RB-1	C: The Complete Reference, Herbert Schildt, 4th Edition, 2000, McGraw Hill
RB-2	Learn C Programming: A Beginner's Guide to Learning C Programming the Easy and Disciplined Way, Szuhay, Jeff., EBook, 2020, Packt Publishing

Other Reading (OR):

OR-1	https://www.programiz.com/c-programming/examples
OR-2	https://beginnersbook.com/2015/02/simple-c-programs/
OR-3	https://www.tutorialgateway.org/c-programming-examples/
OR-4	https://codescracker.com/c/program/c-programming-examples.htm
OR-5	https://www.unf.edu/~wkloster/2220/ppts/cprogramming_tutorial.pdf

Relevant Website (RW):

RW-1	https://www.tutorialspoint.com/cprogramming/index.htm
RW-2	https://www.javatpoint.com/c-programming-language-tutorial
RW-3	https://www.w3schools.com/c/
RW-4	https://www.geeksforgeeks.org/c-language-set-1-introduction/

Video Links (VL):

VL-1	https://www.youtube.com/watch?v=mN7H_0cINGA
VL-2	https://www.youtube.com/watch?v=lx578mIRZv4
VL-3	https://www.youtube.com/watch?v=C1I71RBApTA
VL-4	https://www.coursera.org/specializations/computational-thinking-c-programming
VL-5	https://www.coursera.org/specializations/c-programming
VL-6	https://nptel.ac.in/courses/106104128
VL-7	https://www.youtube.com/watch?v=Z_0xXmOgYtY
VL-8	https://www.youtube.com/watch?v=t9WKOcRB63Q&list=PLJ5C_6qdAvBFzL9su5J-FX8x80BMhkPy1

MOOC Courses (MC):

Sr.No.	Platform	Topic	CO	Link	Duration
MC-1	LinkedIn Learning	C Essential Training	CO1	https://www.linkedin.com/learning/c-essential-training/getting-everything-ready?autoSkip=true&autoplay=true&resume=false&u=92695330	2hr 51 min
MC-2	LinkedIn Learning	Programming foundations in C	CO1 CO3	https://www.linkedin.com/learning/learning-c-5/programming-foundations-in-c?autoplay=true&contextUrn=urn%3Ali%3AlyndaLearningPath%3A1~AAAAAAWGayl%3D1723626&u=92695330	3hr 17Min
MC-3	LinkedIn Learning	Write better secure C code	CO3 CO4	https://www.linkedin.com/learning/secure-coding-in-c/write-better-secure-c-code?autoplay=true&u=92695330	1hr 19 min
MC-4	LinkedIn Learning	Master C Language Pointers	CO3 CO4	https://www.linkedin.com/learning/master-c-language-pointers/conquer-your-pointer-fears?autoplay=true&u=92695330	1hr 33 min
MC-5	LinkedIn Learning	Advanced C Programming	CO1 CO3 CO4 CO5	https://www.linkedin.com/learning/advanced-c-programming/go-beyond-the-basics-of-c-programming?autoplay=true&u=92695330	1hr 23 min
Total MOOC Course integration with Certification					11 hours

Session-wise Planning:

Module	Session	Topic	TB/RB/OR/RW/ VL	Pedagogy/ Activity planned	CO	Mode of delivery
Bride Course	Zero Lecture	Introduction to computer- based problem solving in C	VL-1 to VL-8	Blended Learning Approach		synchronous- PPT
I	1	Program design and implementation issues.	TB-1 RW1 MC-1 VL-1,VL-6	Blended Learning	CO1	Synchronous-PPT
	2	Prepare flowcharts		Blended Learning		Interactive Lecture
	3	algorithms		Blended learning		Synchronous-PPT
	4	top down design		Blended Learning		Synchronous-PPT
	6	Programming environment		Blended Learning		Synchronous-PPT
	7	Machine language, assembly language, high level languages,		Blended Learning		Synchronous-PPT Asynchronous- Video
	8	Assemblers, Compilers, Interpreters.		Blended Learning		Open Discussion/Debate
	9	Overview of C, Data Types,		Interactive Discussion with Demo		Synchronous- PPT
	10	Type modifiers and storage class specifiers for data types,	PPT Presentation & Classroom interaction	PPT Presentation & Classroom interaction		Synchronous- PPT
	11	Constants & Variables, Operators, ?Operator, &operator		PPT Presentation & Classroom interaction		Synchronous- PPT
	12	operator, & Expressions,		Blended Learning		Synchronous- PPT
II	13	Branching and	TB-1 RW1 MC-2, VL-6	Blended Learning		Synchronous- PPT

		Looping-			CO2	
	14	IF statement, IF..else statement, Nesting of if..Else statement		Blended Learning		Synchronous-PPT
	15	Else if ladder, Switch Statement, The? : Operator		PPT Presentation & Classroom interaction		Synchronous - PPT
	16	Go To statement, While, Do while,		Blended Learning		Synchronous - PPT
	17	For, break and continue statements		PPT Presentation & Classroom interaction		Synchronous - PPT
	18	Arrays with One, Two dimensional		Blended Learning		Synchronous-PPT
	19	multidimensional arrays.		Blended Learning		Synchronous-PPT
	20	Introduction to strings.		Blended Learning	CO2	Synchronous-PPT
	21	Type casting		PPT Presentation & Classroom interaction		Synchronous-PPT
	22	type Conversion.		PPT Presentation & Classroom interaction		Synchronous-PPT
	23	scanf(), printf(), gets(), puts()		PPT Presentation & Demonstration		Synchronous-PPT
	24	getch() and putch().		Blended Learning		synchronous- PPT
III	25	Introduction to Library Functions	TB-1 RW1 MC-3 VL-2,VL-6	PPT Presentation & Demonstration	CO3	Synchronous-PPT
	26	Introduction to User defined functions		PPT Presentation & Classroom interaction		Synchronous-PPT
	27	Call by value		PPT Presentation &		Synchronous-PPT

				Demonstration		
	28	Call by reference,		PPT Presentation & Classroom interaction	CO3	Synchronous-PPT
	29	calling functions with arrays		PPT Presentation & Demonstration		Synchronous-PPT
	30	argc and argv,		Blended Learning		Synchronous-PPT
	31	Recursion- basic concepts, ex-towers of Hanoi.		PPT Presentation & Demonstration		Synchronous-PPT
	32	Activity Work		Blended Mode with Interactive Assessment		Synchronous-PPT &
IV	33	Introduction to pointers	TB-1 ,TB-2 VL-6,VL-7 VL-3, RW-2	Blended Learning	CO3	Synchronous-
	34	Declaration and Initialization		Blended Learning	CO3	Synchronous-PPT
	35	The & and * operator		Demonstration with Classroom Interaction		Synchronous-PPT
	36	pointer expression, arithmetic,		Blended Learning	CO3	Asynchronous- PPT
	37	comparison, malloc vs calloc		Blended Learning	CO3	Synchronous-PPT
	38	arrays of pointers		PPT Presentation & Classroom interaction	CO4	Synchronous-PPT Lecture with interaction
	39-40	arrays of pointers, pointers to pointers, pointers to functions		PPT Presentation & Demonstration		Synchronous-PPT
	41-43	function returning pointers. Basics, declaring, referencing structure elements,		PPT Presentation & Classroom Interaction		Synchronous-PPT
	44-46	array of structures, passing structures to functions, structure pointers,		PPT Presentation & Problems		Synchronous-PPT

		arrays and structures within structures.				
	47-48	Declaration, uses, enumerated data-types, typedef		PPT Presentation & Classroom Interaction		Synchronous-PPT &
V	49-50	The file pointer, file accessing functions,	TB-1 RW-3 VL-6,VL-7, VL-8	PPT Presentation & Classroom interaction	CO5	Synchronous-PPT Asynchronous-Video
	51-53	fopen, fclose, puc, getc, fprintf		PPT Presentation & Classroom interaction		Synchronous-PPT
	54-55	#define, #include, #undef, Conditional compilation directives.		PPT Presentation & Classroom interaction		Synchronous-PPT
	56-57	Header files, string functions, mathematical functions, Date and Time functions		PPT Presentation, Demonstration & Classroom interaction		Synchronous-PPT
	58-60	Revision of topics and doubt clarifications		PPT Presentation & Classroom interaction		Synchronous-PPT Asynchronous-Video Report: Examine the activities

Assessment Scheme: CA: UE – 50:50

Sl. No.	Assessment Instrument	Formative/ Summative	Frequency/Stages	Weightage (%)	CO
1.	Activity-1	Formative	1	15	CO1,CO2,CO3,C O4
2.	Activity-2		1	15	CO2,CO3,CO4,C O5
3.	Class Participation	Formative	-	5	
4.	Periodic Test	Formative	3	15	CO1-CO5
5.	End Semester Examination	Summative	1	50	CO1-CO5
Total				100	

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